

Fibrillary Glomerulonephritis

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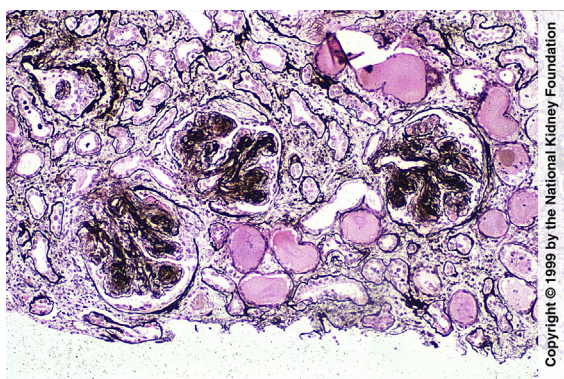


Fig 1. Fibrillary glomerulonephritis has varying appearances by light microscopy, with mild to extensive mesangial proliferative and membranoproliferative appearance being the most common. In this case, there was a distinct membranoproliferative and even nodular appearance (Jones silver stain; original magnification $\times 100$).

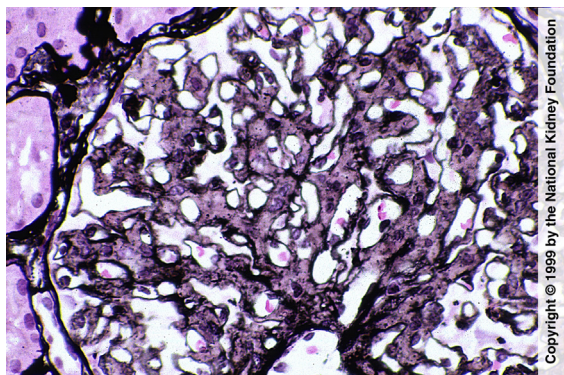


Fig 2. Fibrillary glomerulonephritis may have varying degrees of mesangial proliferation or membranoproliferative pattern by light microscopy. In this case, there was moderate mesangial proliferation and occasional basement membrane splitting. Additional studies confirmed the diagnosis of fibrillary glomerulonephritis with typical electron microscopy appearance, negative Congo red stain, and IgG staining by immunofluorescence (Jones silver stain; original magnification $\times 400$). See other figures.

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Some cases kindly shared by Dr Robert G. Horn, Private Laboratory for Renal Biopsy Pathology, Nashville, TN; Medical Photographer: Brent Weedman; with assistance from Kim Soletz, MD, of the National Kidney Foundation's cyberNephrology™ Team.

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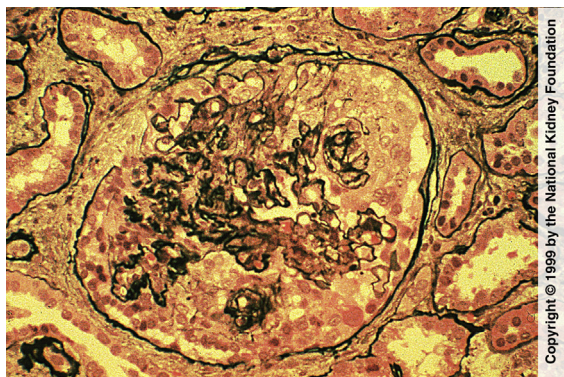


Fig 3. In approximately one third of cases seen in our series of fibrillary glomerulonephritis, crescents were present, as in this case. In some of these, confluent deposits caused immunofluorescence to have a pseudolinear appearance, which could give rise to diagnostic confusion until electron microscopy is examined (Jones silver stain; original magnification $\times 200$).

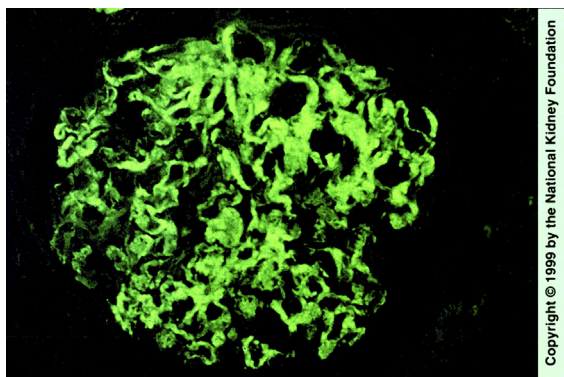


Fig 4. Immunofluorescence studies in fibrillary glomerulonephritis are striking and characteristic with smudgy positivity in mesangial and capillary loop areas. The mesangial staining most commonly is more extensive than the capillary loop staining, and the typical smudginess allows suspicion of fibrillary glomerulonephritis. The most common staining is IgG, with IgG4 subtype typically being most prominent (anti-IgG immunofluorescence; original magnification $\times 400$).

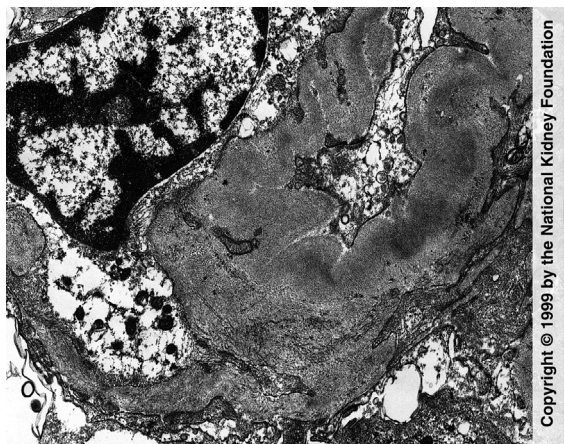
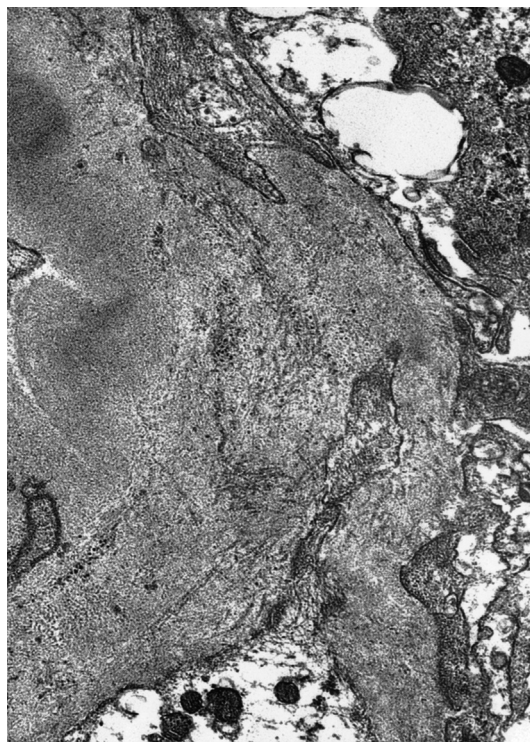
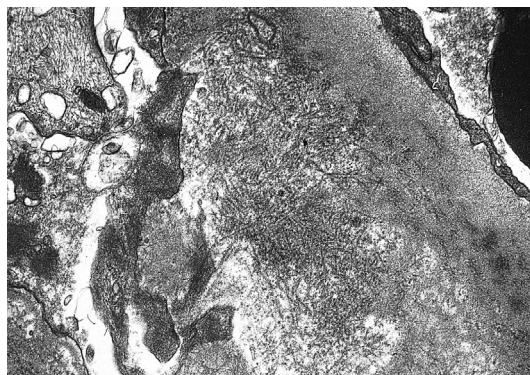


Fig 5. The diagnosis of fibrillary glomerulonephritis is ultimately made by electron microscopy, which shows randomly arranged fibrils in mesangial and capillary loop areas with an intramembranous, subepithelial, and/or subendothelial location. Negative Congo red stains are necessary to specifically exclude the possibility of amyloid (transmission electron microscopy; original magnification $\times 25,625$).



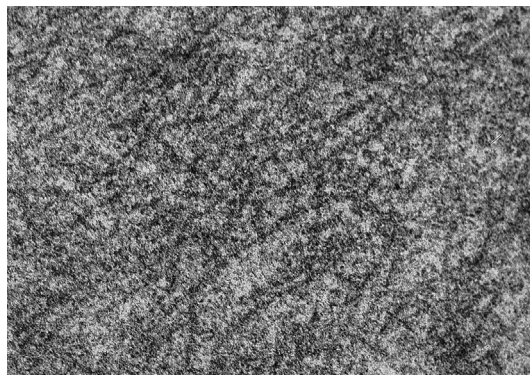
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Fig 6. Higher power shows randomly oriented fibrils in the mesangium and glomerular basement membrane in fibrillary glomerulonephritis. These fibrils tend to be slightly larger than those in amyloid, although there is some overlap so that fibril size alone cannot be used to definitively distinguish this from amyloid. A negative Congo red stain should be done to rule out amyloid deposits (transmission electron microscopy; original magnification $\times 51,250$).



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Fig 7. Distinct subepithelial localization of fibrillary deposits in a case of fibrillary glomerulonephritis. In this case, immunofluorescence and light microscopic studies gave the impression of a membranous glomerulonephritis, illustrating the utility and need for electron microscopy to definitively evaluate renal biopsies. The fibrils are randomly oriented and somewhat larger than in amyloid, 12 to 15 nm, although there is some overlap so that size alone cannot be used to definitively distinguish this from amyloid deposits (transmission electron microscopy; original magnification $\times 62,000$).



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Fig 8. High-power view of fibrils in fibrillary glomerulonephritis, illustrating slightly coarser diameter than in amyloid. The fibrils are randomly arranged in the loose background. A negative Congo red stain is necessary to definitively rule out amyloid deposits, although fibrillary glomerulonephritis can be suspected from the characteristic immunofluorescence microscopy and the slightly larger fibril size (transmission electron microscopy; original magnification $\times 98,000$). See other figures.